

# Problem Set 2, COSC330/PHYS306

## Sections 2-8, 2-11, and 3-2 through 3-6

Prof. Jordan C. Hanson

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### 1 Chapter 2

1. [Section 2-8, Exercise 38] Convert each binary number to hexadecimal:

- (a) 1110    **E**
- (b) 10    **2**
- (c) 10111    **17**
- (d) 10100110    **A6**
- (e) 1111110000    **3F0**
- (f) 100110000010    **982**

2. [Section 2-8, Exercise 39] Convert each hexadecimal number to decimal:

- (a)  $23_{16}$     **35**
- (b)  $92_{16}$     **146**
- (c)  $1A_{16}$     **26**
- (d)  $8D_{16}$     **141**
- (e)  $F3_{16}$     **243**
- (f)  $EB_{16}$     **235**
- (g)  $5C2_{16}$     **1474**
- (h)  $700_{16}$     **1792**

3. [Section 2-11, Exercise 56] Convert each binary number to Gray code:

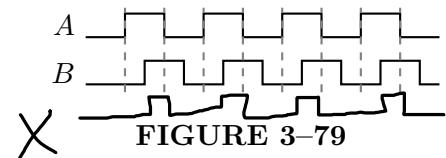
- (a) 11011    **10110**
- (b) 1001010    **1101111**
- (c) 1111011101110    **100011011001**

4. [Section 2-11, Exercise 57] Convert each Gray code to binary:

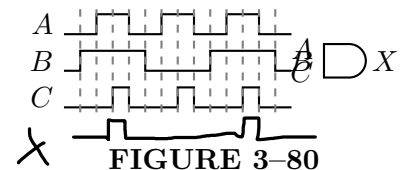
- (a) 1010    **1100**
- (b) 00010    **00011**
- (c) 11000010001    **10000011110**

### 2 Chapter 3

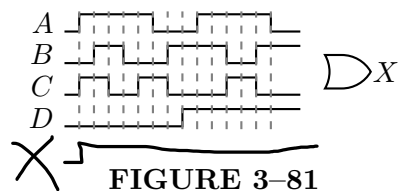
5. [Section 3-2, Exercise 6] Repeat Problem 5 for the waveforms in Figure 3-79.



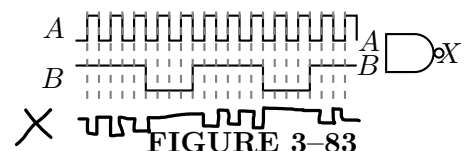
6. [Section 3-2, Exercise 7] The input waveforms applied to a 3-input AND gate are as indicated in Figure 3-80. Show the output waveform in proper relation to the inputs with a timing diagram.



7. [Section 3-3, Exercise 12] Repeat Problem 8 for a 4-input OR gate.



8. [Section 3-4, Exercise 17] For the set of input waveforms in Figure 3-83, determine the output for the gate shown and draw the timing diagram.



9. [Section 3-4, Exercise 18] Determine the gate output for the input waveforms in Figure 3-84 and draw the timing diagram.

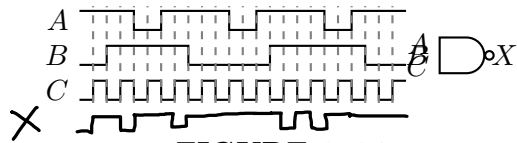


FIGURE 3-84

10. [Section 3-5, Exercise 22] Determine the output waveform in Figure 3-87 and draw the timing diagram.



FIGURE 3-87

11. [Section 3-5, Exercise 24] The NAND and the negative-OR symbols represent equivalent operations, but they are functionally different. For the NOR symbol, look for at least one HIGH on the inputs to give a LOW on the output. For the negative-AND, look for two LOWs on the inputs to give a HIGH output. Using these two functional points of view, show that both gates in Figure 3-88 will produce the same output for the given inputs.

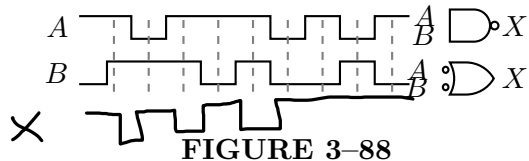


FIGURE 3-88

12. [Section 3-6, Exercise 26] Repeat Problem 17 for an exclusive-OR gate. Output HIGH only when one is on not both
13. [Section 3-6, Exercise 27] Repeat Problem 17 for an exclusive-NOR gate.

Output HIGH only when both inputs are the same. like an XAND Gate